

Lab 0 - Rubber Stoppers

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1 Purpose

The purpose of this lab is to determine the density of two sizes of rubber stoppers and to write up a complete lab report using proper format.

2 Introduction

Density is a physical property that describes the relationship between the mass of a material and its volume. This property may be used to identify a material and can be used to compare two different materials. Density is the ratio calculated using $\rho = \frac{m}{V}$, where ρ is the density, m is the mass, and V is the volume.

This lab will use the displacement of water to determine the volume of the rubber stoppers, which in addition to the mass taken on a mass scale, will then be used to calculate the densities of two rubber stoppers. This lab will explore two different methods of finding the volume of the rubber stoppers, one for each size, due to the lab's constraint of not being able to use a larger graduated cylinder. The smaller rubber stopper will be measured using a graduated cylinder, while the larger rubber stopper will be measured using a beaker and a watch glass, where the change in mass of the watch glass will then be measured to determine the volume.

3 Materials

- Graduated cylinder (200 mL)
- Two rubber stoppers (small & large)
- Beaker (50 mL)
- Watch glass
- Water
- Mass scale
- Dropper

4 Procedure

4.1 Part 1: Density of Small Rubber Stopper

<u>Materials/Equipment</u>	<u>Actions Taken</u>
Mass Scale	Measured mass of small rubber stopper
Graduated Cylinder	Filled graduated cylinder about half-way with water Recorded initial water level
Small stopper	Dropped small rubber stopper into water stirred to remove bubbles Recorded final water level

4.2 Part 2: Density of Large Rubber Stopper

<u>Materials/Equipment</u>	<u>Actions Taken</u>
Mass Scale	Measured mass of large rubber stopper
	Measured initial mass of watch glass
Beaker (50 mL)	Filled beaker most of the way with water
	Lefted enough space to move without spilling
Watch Glass	
Beaker (50 mL)	Placed beaker (50 mL) on watch glass
Dropper	Used dropper to add water to beaker (50 mL) until full
Large Stopper	Slowly lowered large rubber stopper into beaker (50 mL)
Dropper	Removed water from beaker (50 mL) to prevent spillage
Beaker (50 mL)	Removed beaker (50 mL) from watch glass
Mass Scale	Measured final mass of watch glass

5 Observations

Table 1: Observations of Rubber Stoppers

Rubber Stopper Size	Mass (g)	Initial Measurement	Final Measurement
Small	4.33g	15.1 mL	16.8 mL
Large	24.25g	25.59g	41.07g

6 Analysis

6.1 Density of Small Rubber Stopper

$$\begin{aligned}\text{Volume} &= \text{Final} - \text{Initial} \\ &= 16.8 \text{ mL} - 15.1 \text{ mL} \\ &= 1.7 \text{ mL} \\ &= 1.7 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Density} &= \frac{\text{Mass}}{\text{Volume}} \\ &= \frac{4.33 \text{ g}}{1.7 \text{ cm}^3} \\ &\approx 2.55 \text{ g/cm}^3 \\ &\approx 2.6 \text{ g/cm}^3\end{aligned}$$

6.2 Density of Large Rubber Stopper

Assuming the density of water is 1 g/mL.

$$\begin{aligned}\text{Volume} &= (\text{Final Mass} - \text{Initial Mass}) \times 1 \text{ g/mL} \\ &= 41.07 \text{ g} - 25.59 \text{ g} \\ &= 15.48 \text{ g} \\ &= 15.48 \text{ mL} \\ &= 15.48 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Density} &= \frac{\text{Mass}}{\text{Volume}} \\ &= \frac{24.25 \text{ g}}{15.48 \text{ cm}^3} \\ &\approx 1.567 \text{ g/cm}^3\end{aligned}$$

7 Discussion

1.

$$\begin{aligned}\text{Percentage Error} &= \left| \frac{\text{Experimental Value} - \text{Theoretical Value}}{\text{Theoretical Value}} \right| \times 100\% \\ &= \left| \frac{1.567 - 2.55}{2.55} \right| \times 100\% \\ &\approx 38.5\%\end{aligned}$$

$\therefore$ the percentage error for the method from Part 2 is approximately 38.5%.

2. In order to find the percentage composition of copper and zinc in a brass sample:

- Take the mass of the brass sample using a mass scale.
- The volume of the brass sample would be determined using water displacement in a graduated cylinder.
- The density would be calculated using the formula $\rho = \frac{m}{V}$.
- The percentage composition of copper and zinc in the brass sample could be determined by using a system of equations with the known densities of pure copper and pure zinc.